CS4530 Final Project: “FakeStackOverflow”

Group 801: Giacomo Mantovanelli, Siming Fu, Krish Angra

https://fakestackoverflow-801.onrender.com/
https://github.com/neu-cs4530/fall25-project-801

Our Feature: Security, Cosmetics, and Blocking

The base implementation of FakeStackOverflow simulated a forum website for users to post questions and receive answers from others, alike the real StackOverflow website. Although the website held a solid foundation for a forum website, we gathered and saw more potential pain points for users: Can the user trust the website is safe for their information and that their accounts will be safe against predictable attacks? Can the user expect some sort of reward for their participation in the community as a whole? Can the users have control over who they wish to interact with?

In order to address these issues, we built, tested, and deployed a major quality of life update, including **security features**, **cosmetics & shop**, **and blocking**.

- Passwords are never stored plaintext but rather Salted + Hashed
- JWT Signed Cookies are used for user authentication
- Aimed to further incentivize users community interactions, we’ve built a system of personal points and cosmetics. This way, users are able to earn points by providing quality answers to questions, and playing mini-games with other users. Afterwards, they are able to purchase public cosmetics for others to see. Additionally, we have also added another type of minigame for users to play. We hope this will encourage users to consistently engage in the FakeStackOverflow community.
- Users can block other users, preventing message delivery

Our Implementation: Technology Stack & Design

Overall we stuck with the layered architecture. We also overall used the push/pull pattern for all requests that didn’t need real time updates. For Real time updates we used sockets. This approach allowed us to keep a mostly stateless server while allowing for instant updates on our messaging features.

The cosmetic system was implemented closely following the structure provided by other services, as a server-client https architecture connecting the front and back end. Cosmetic has its own table in the database, and it is a strictly one-to-one relationship with the user table. Each user’s cosmetic entry contains their current number of points, and two list of enum type, representing the cosmetics that they own and currently equipped. The enum type itself is defined as a collective ‘super-type’ of different cosmetic items, so that future expansion of cosmetics can be systematic and simple.
Likewise, the Connect Four minigame also closely follows the existing Nim structure, and uses pre-existing frontend and backend game services.

Future Work

Expanding the server authentication to log ip addresses and other device identifiable information could allow us to increase our user security by adding extra layers that verify the same cookie isn’t used on multiple devices.

By expanding on the cosmetic types and adjusting the front end shop, the cosmetic system intentionally leaves behind possibility of further variations of different cosmetic systems to offer. Similarly the game system also can be expanded upon to offer other, more engaging minigames.
Technically, there are still clean-up work to be done on cosmetic services. Chronologically it was one of the last finished and definitively could use more refactoring work. For example, currently a bug that we did not have time to fix is that by cancelling upvote and re-upvoting, one user can generate points repeatedly for another user.

We could expand community feeling to the communities by adding more community only things. For example we could make it so some game invites are only available to community members. Or some questions…

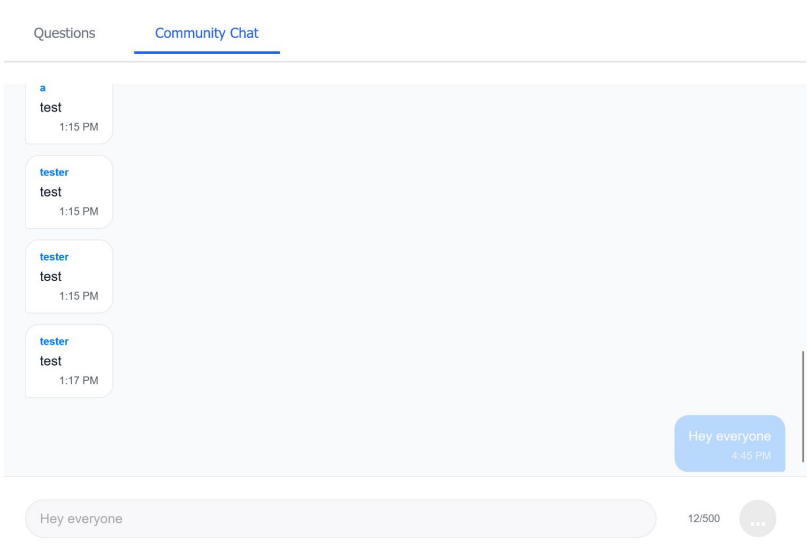
In the future we could expand the blocking feature to limit ALL interactions from the blocked user to the blockee. Including answering questions, playing games, interacting with questions…

Images

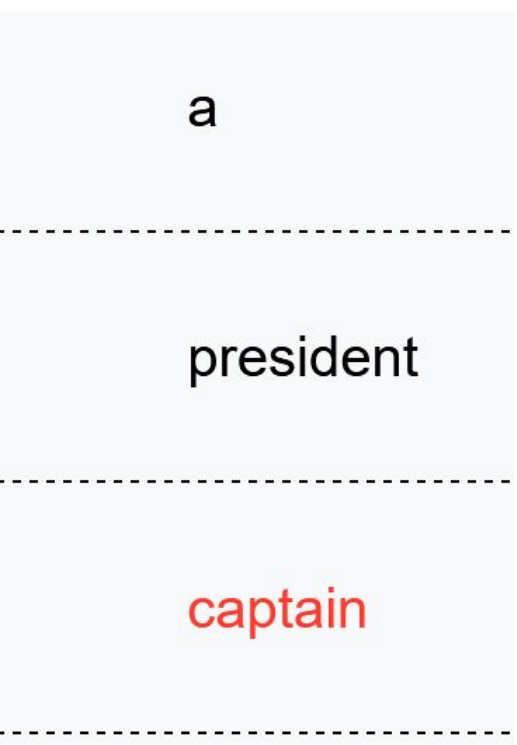
Cookie Issued and stored in browser

Name	Value	Domain	Path
authToken	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VybmFtZSI6ImNhcHRhaW4iLCJ2ZXJzaW9uIjoxLCJleHRlbmRlZ...	fakestackoverflow-801.onren...	/

Community page has a community chat



Captain user purchased and equipped a red username cosmetic.



Cosmetics Page

